Algebra worksheet

Question 1. Solve 2x + 7 = 19. Give the value of x.

Answer: 6

Question 2. Expand 3(x + 4).

Answer: $3x + 12$

AI-generated worked solutions

Written by an AI and may contain mistakes. Answers marked as not confidently solved or not checked need a careful look.

Question 1

Solve 2x + 7 = 19. Give the value of x.

Checked by calculation

Subtract 7 from both sides to isolate the term with x.

$$2x = 19 - 7$$

Simplify the right hand side.

$$2x = 12$$

Divide both sides by 2 to find x.

$$x = \frac{12}{2}$$

Calculate the final value.

$$x = 6$$

Answer: 6

To solve for x, subtract 7 from both sides of the equation and then divide by 2.

Question 2

Expand 3(x + 4).

Checked by calculation

Multiply 3 by x.

$$3 \times x = 3x$$

Multiply 3 by 4.

$$3 \times 4 = 12$$

Combine the results.

$$3x + 12$$

Answer: $3x + 12$

To expand the expression, multiply the term outside the brackets by each term inside the brackets.